

Traversing The SOLAR PATH OF INDIA



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India, with an ambitious goal of installing 100GW of solar energy capacity under Jawaharlal Nehru National Solar Mission (JNNSM) by 2022, is expecting a call for huge capital investment—to the tune of ₹ 6 trillion. In addition, solar power projects requiring investments of about ₹ 900 billion are planned to be bundled with thermal power projects. Typical investors would include large public sector undertakings and independent power producers. State governments have also come out with state-specific solar policies to promote addition of solar capacity. As part of India's ambitious goal, total solar power installed capacity in the country has crossed 5GW in 2015, including 4.7GW of utility-scale solar and 525MW of rooftop solar power capacity.

The growth drivers

One of the key reasons why solar industry is growing in India is the fact that the country itself is developing. Good power supply is one of the key enablers of development these days. India's power requirement is expected to grow by more than five per cent annually over the next decade, while supply is projected to fall short by 3.6 per cent. Here, the alarming fact is that traditional energy sources are not

enough to match the pace of growing demand.

So people are coming forward to look for alternatives, and solar fits the bill. Major factors boosting the current and future growth in the solar energy segment are summarised in Table I.

On the way to Make in India

Growth prospects along with the proposed five-fold increase in JNNSM targets will definitely boost solar manufacturing industry in the country. This, in turn, will help in the creation of technology hubs for manufacturing broadly in sync with Make in India initiative.

Apart from local players, global solar manufacturers also have an eye on India as one of their preferred destinations for solar energy investment. According to industry sources, there are a host of global renewable energy firms that have shown an interest in investing here. India has made it clear that it will encourage global as well as local solar manufacturers to make in India and is offering many incentives to these firms to start production here. Recently, we have seen a flurry of global investors investing in states like New Delhi, Tamil Nadu and Kerala. The numbers are expected to increase substantially in the coming months.

However, numerous challenges need to be addressed to make Make in India initiative a reality for solar manufacturing. Off-grid solar power systems face major challenges like high capex investment requirements, inefficient subsidy disbursement processes, no minimum guarantee of return on investment and so on. Moreover, it is not feasible to render any generic installation approach due to varying consumer locations and preferences. Thus, increasing solar capacity in the country requires planning and innovation to get rid of those challenges.

Challenges to be addressed

Multiple decentralised schemes are available from different ministries to facilitate solar electrification. These need to be integrated or linked to provide clear directives to the indus-

TABLE I
SOLAR INDUSTRY GROWTH FACTORS IN INDIA

Demand side	Supply side
• Rising energy needs in the country • Persistent energy shortage in the country with average demand-supply gap revolving around 12 per cent of total power supply • Future increase in per capita electricity consumption up to around 1900kWh by 2032 from the current consumption of more than 1000kWh (as per the power ministry forecasts) • Favourable policy measures such as JNNSM aimed at encouraging investment in the solar energy sector • Increasing public awareness about issues such as energy scarcity and environmental sustainability fuelling the demand for eco-friendly power, hinting at growth opportunities for solar power	• Currently, power generation in India is heavily dependent on non-renewable natural resources such as coal and diesel, resulting in fast depletion of those resources. It is compelling the government and power-generation companies to consider renewable energy sources, especially solar power • The favorable environment created by government through subsidy schemes and policies encourage power-generation companies to invest in solar sector and thus promoting growth • Huge demand for electricity in rural areas lacking grid connectivity and abundant availability of sun in India throughout the year